5G LMF Description

NLS

User Guide

17/1553-HSC 901 1780 Uen G

Contents

1 Introduction

The Location Management Function (LMF) deployed in 5G core (5GC) network supports control plane positioning for subscribers in the standalone New Radio (NR) network.

2 Technical Description

2.1 LMF Overview

The NLS implements the LMF Network Function (NF) defined by 3GPP. This section gives an overview of the 3GPP Location Services (LCS) architecture and the interfaces related to the LMF.

The 5GC architecture is service-based. The system functionality in Service-Based Architecture (SBA) is achieved with a set of NFs that provide service access to other authorized NFs. Authorization can be achieved by enabling OAuth 2.0 support. An NF is a component of the network infrastructure that provides a specific functional behavior such as location determination. An NF is made up of NF services.

The interactions between NFs are represented in the following ways:

— Reference point representation

— Service-based representation

Figure 1 shows the reference point representation of 5G LCS defined by *3GPP TS 23.501*, where the interaction between NFs is described by point-to-point reference points.

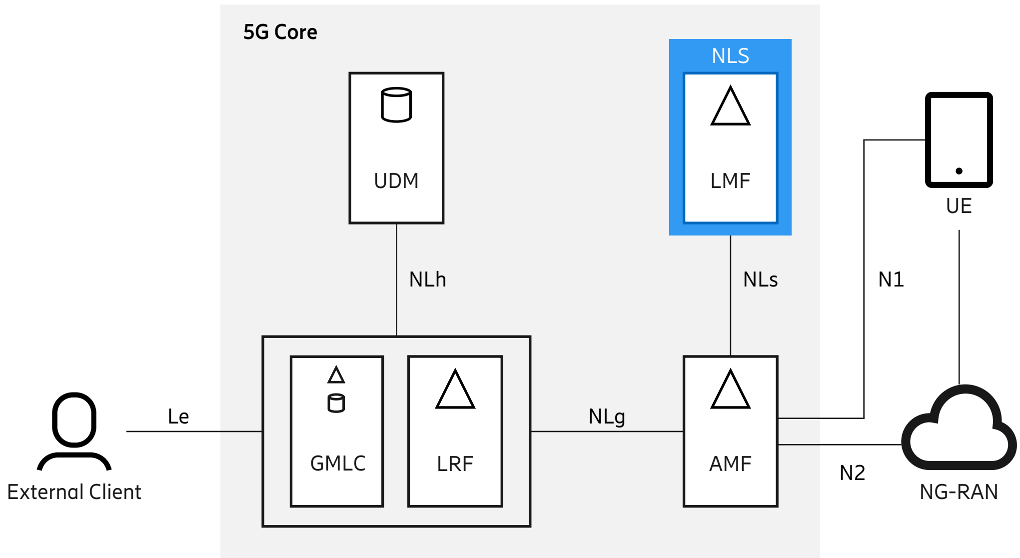

Figure 1 System Architecture in 5GC

The following reference points support LCS in 5GC:

— N1: Reference point between a UE and AMF through NAS

— N2: Reference point between NG-RAN and AMF

— Le: Reference point between a GMLC and an LCS client

The following reference points are realized by Service-Based Interfaces (SBI):

— NLg: Reference point between a GMLC and AMF

— NLs: Reference point between AMF and LMF

— NLh: Reference point between a GMLC and UDM

Figure 2 The following shows the service-based representation of the 5G LCS architecture, where the network functions are interconnected through SBIs.

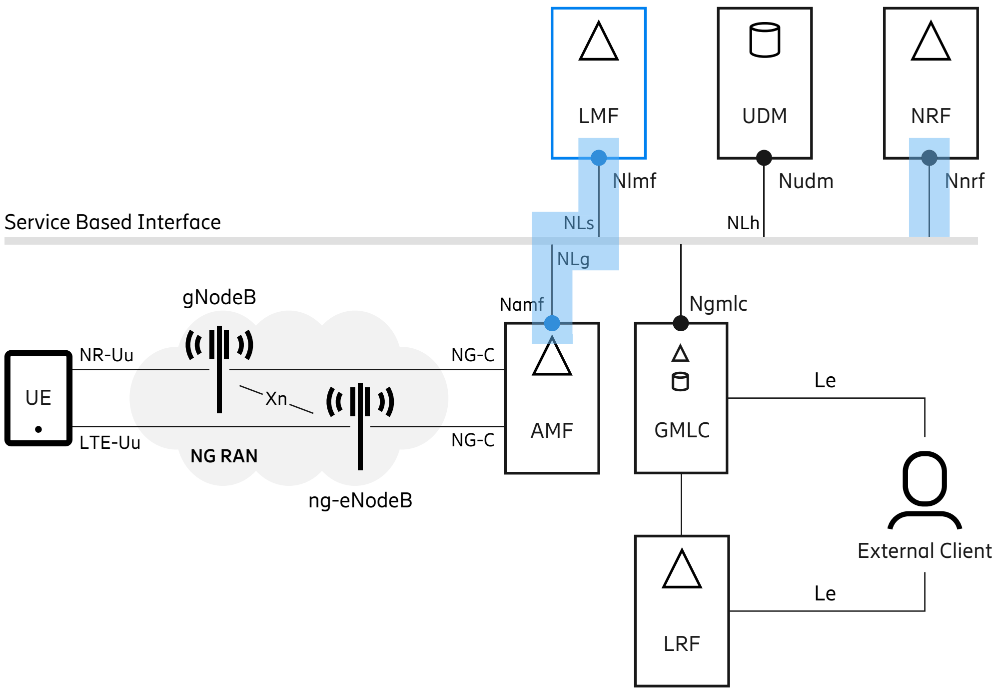

Figure 2 5G LCS Architecture

System Architecture inThe following SBIs are available for LCS:

— Ngmlc: SBI exhibited by GMLC

— Nlmf: SBI exhibited by LMF

— Namf: SBI exhibited by AMF

— Nudm: SBI exhibited by UDM

— Nnrf: SBI exhibited by NRF

The LMF supports the Nlmf_location and the Nlmf_broadcast services.

The Nlmf_location service enables the AMF to request the location of a target UE. The Nlmf location service uses the Namf interface provided by the AMF for transferring positioning information to the UE or the NG-RAN.

The Nlmf_broadcast service enables the LMF to receive RTK broadcast-related feedback (Assistance Information Feedback messages) from the gNodeBs through the AMF. The RTK broadcast is not associated with a UE location session. The broadcast is used in an AMF to send network assistance data to an NG-RAN node. The NG-RAN node broadcasts the assistance data to the target UEs. LMF supports AMF in distributing ciphering keys to UEs. UEs can decipher RTK broadcast assistance data, ciphered by the LMF, using the distributed ciphering keys. The ciphering procedure is not associated with a UE location session.

For further information regarding the RTK broadcast feature of the LFM, see the related *RTK Broadcast* document.

The LMF also interacts with the Network Repository Function (NRF) through the Nnrf interface for NF service framework functions. The LMF supports sending the User-Agent header on the Namf and Nnrf interfaces with a configurable value. For more information, see the *PPLMF Parameters on Server Level* table in the section mentioned in the Related Information .

According to the 3GPP standard (3GPP 23.501) communication models, there can be direct or indirect communication with external NFs. In the latter case, the communication is done through the Service Communication Proxy (SCP). LMF is prepared to perform Nlmf_location service-related communication with AMF indirectly.

Note: For indirect communication, LMF supports only model C, not model D.

LMF does not support the delegation of NF producer selection to the SCP.

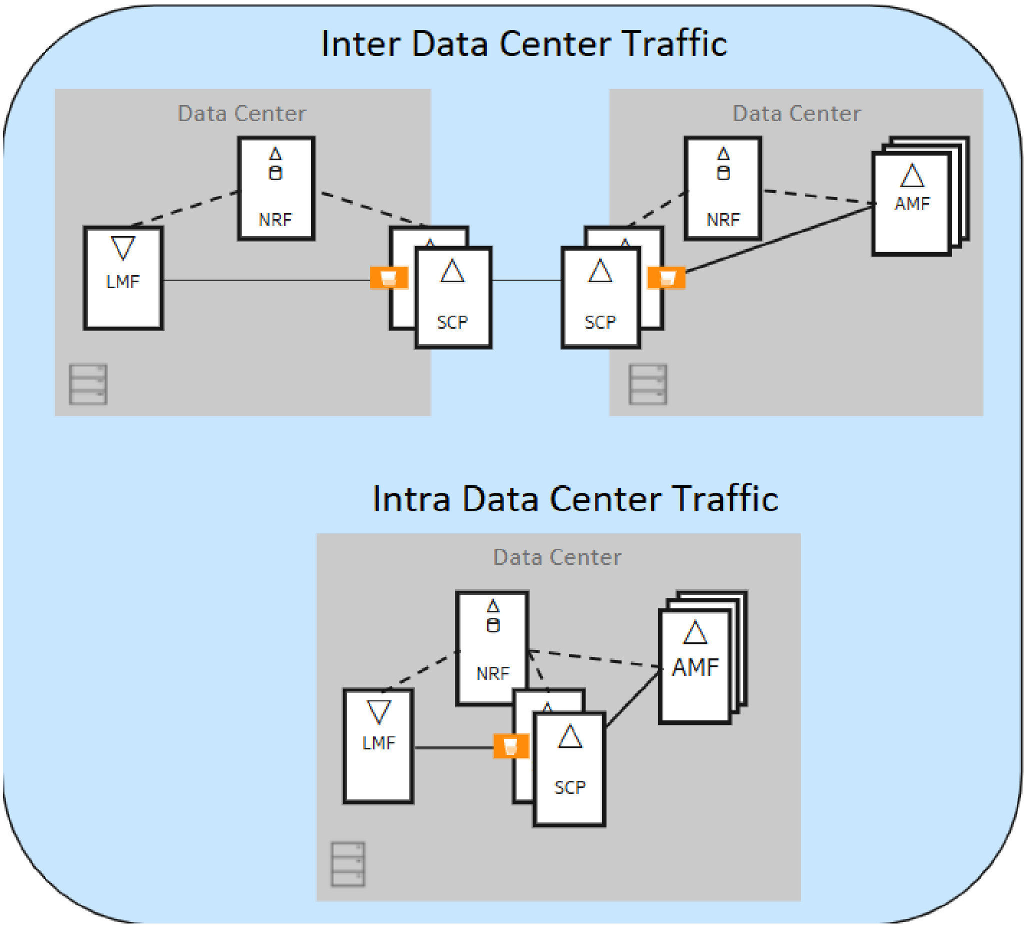

Figure 3 SCP Deployment between NF Producer and NF Consumer

—— RELATED INFORMATION ——

Parameter List (NLS)

RTK Broadcast

2.1.1 Feature Overview

The NLS implements the LMF entity for the 5G LCS architecture and the procedures defined by 3GPP. The LMF entity is co-located on the NLS server and shares internal supporting functions with the E-SMLC, the SAS, and the SMLCs. The internal supporting functions include the GNSS reference network, database, and O&M functions.

The LMF server receives `DetermineLocation` request messages from the AMF, and determines the location of the target UE. For both UE-based and UE-assisted positioning methods, the LMF exchanges LPP, LPPe, and NR Positioning Protocol A (NRPPa) messages with the target UE through the AMF communication service. The LMF returns the location estimate in the Location Response message.

The LMF supports the following positioning methods:

— Cell ID
— NR ECID (NR Enhanced Cell ID)
    • Outdoor Synchronization Signal Reference Signal Received Power (SS-RSRP) based positioning
    • Indoor Radio Dot System (RDS) precise positioning
— NR AECID (NR Adaptive Enhanced Cell ID)
— UE-assisted A-GNSS
— UE-based A-GNSS
— Uncompensated Barometric Pressure (UBP)
— Device Based Hybrid (DBH)
— Civic Address

The LMF provides the fixed XY data, which overrides the NR cell data during the following positionings:

— CID
— NR ECID
— NR AECID
— A-GNSS

For A-GNSS, it is used only as the reference location for assistance data calculation.

The LMF supports a flexible positioning method control mechanism based on the UE's positioning capabilities, the positioning method control configuration, and the parameters included in the `Location Request` message. For NR positioning, the positioning method control mechanism is similar to that in GSM, WCDMA, or LTE, with the following differences:

— The configurations of the positioning methods are supported only on global level.
— The positioning methods are defined as Primary, PrimaryFallback, PrimaryFallback2, Secondary, SecondaryFallback, and SecondaryFallback2.
— The LMF does not support dual technology and the hybrid methods.

Depending on which positioning method is used to generate the location information, the shape of the location estimate is different. The returned shape can be configured according to the original shape and client type. The LCS client can specify a supported shapes list in the `Location Request` message. If the returned shape is not in the supported shape list, the LMF returns a positioning failure message to the AMF.

The LMF can return a location estimate that does not meet the requested horizontal accuracy. For NR, this is controlled by the configuration for each LCS client type. The LMF can also be configured to perform all available positioning methods and to return the location with the best accuracy.

For detailed information, see the Related Information.

The LPP, LPPe, NRPPa information is the following:

— 3GPP LPP is defined in *3GPP 37.355 v15.0.0*

— OMA LPPe is defined in *OMA-TS-LPPe-V1_0-20150414-C*

— NRPPa is defined in *3GPP TS 38.455 v17.2.0*

— Access and Mobility Management Services is defined in *3GPP TS 29.518 v16.6.0*

— Location Management Services is defined in *3GPP TS 29.572 v16.5.0*

— Network Function Authorization is defined in *3GPP TS 29.510 v17.4.0*

Cell ID

The Cell ID (CID) is the basic method to determine the user location based on the coverage area of the serving cell. When the CID positioning method is selected, the LMF queries the cell data in the database to find the cell's geography information based on the NR CID included in the `Location Request` message. The accuracy of the CID positioning method depends on the size of cell's geographic area. The coverage area of the NR cell can be a polygon or ellipse arc, depending on the parameters included in the NR cell data file.

When CID fallback is enabled, CID positioning is performed if all configured positioning methods fail

NR ECID

— Outdoor SS-RSRP based positioning:

  The NR ECID is a positioning method based on the distance of gNodeB and the UE. The positioning method receives a location request and provides a position. For NR ECID positioning methods, the LMF exchanges NRPPa or LPP messages with the gNodeB or the UE, respectively, through the AMF communication service. The LMF returns the location estimate in the

Location Response message. The NR ECID can only be used in 5G Option 2 environments.

— Indoor RDS precise positioning:

  RDS positioning is based on uplink Time Difference of Arrival measurements (UTDOA) measured between the UE and the radio DOTs. The positioning method receives a location request and provides a position. For RDS positioning methods, the LMF exchanges NRPPa messages with the gNodeB through the AMF communication service. RDS requires periodic report characteristic and Ericsson proprietary extension. The LMF returns the location estimate in the location response message.

NR AECID

The NR AECID is a positioning method based on:

— SS Reference Signal Received Power measurements (SS-RSRP) of the serving and neighbor cells

— Angle of Arrival measurements

— Timing advance measurements

The positioning method receives a location request and provides a position. For NR AECID positioning methods, the LMF exchanges NRPPa messages with the gNodeB or the UE, respectively, through the AMF communication service. The LMF returns the location estimate in the Location Response message. The NR AECID can only be used in 5G Option 2 environments.

A-GNSS

The A-GNSS method uses satellite signal measurements retrieved by systems such as GPS and GLONASS. This method is a highly accurate positioning procedure where the location server, such as the LMF, sends assistance data and enhances positioning performance by shortening the Time To First Fix (TTFF), thereby increasing receiver sensitivity, reducing power consumption, and increasing the accuracy of location calculation.

The LMF supports both UE-assisted A-GNSS and UE-based A-GNSS positioning methods using GPS,GLONASS, or both satellite systems. Autonomous GNSS is supported as a fallback for UE-based A-GNSS and UE-assisted A-GNSS.

The NR ECID Prefix feature calculates the position by using the `NR ECID before A-GNSS` method in case the A-GNSS is set as `Primary` and the NR ECID is set as `Primary fallback`.

If the NR ECID Prefix feature is enabled, the LMF schedules the `NR ECID before AGNSS` method and caches the NR ECID result. After that, the LMF executes the A-GNSS method.

If the A-GNSS method is successful, its result is used as the final result, otherwise the cached NR ECID result is sent to the AMF.

For information on how to activate or deactivate the NR ECID Prefix feature, see *EnableECIDPrefixInAGNSS* in the *PPLMF Parameters on Cluster Level (A-GNSS Positioning)* table in the section mentioned in the Related Information.

For detailed information about A-GNSS positioning, see AGNSS Positioning Technique, ENL.

UBP

The NR UBP solution for LMF is based on LCS-AP and OMA LPPe. LCS-AP messages for UBP are transferred between the LMF and AMF over HTTP2. If UBP over LPPe is supported, the LMF retrieves sensor capability and measurement from the UE over LPPe 1.0, then the LMF returns UBP to the AMF in the location response over HTTP2.

DBH

The LMF supports DBH over LPPe. It is processed only with A-GNSS, not as a standalone positioning. With the DBH method, the UE generates a location estimate from all of its available sources: GNSS, barometric pressure, WLAN, and Bluetooth. The NLS receives the DBH location estimate from UE over LPPe, and selects either the DBH location estimate or the A-GNSS positioning result, then returns it to the AMF.

Civic Address

The LMF supports Civic Address over LPPe. The UE can have the ability to send the civic address over the LPPe based on its calculation. The civic address from the UE is processed and checked against the DBH over LPPe positioning method to make it more reliable.

Note: Civic address is processed only with A-GNSS and DBH, not as a standalone positioning.

Multiple Location Reporting

The LMF supports the Multiple Location Reporting feature. The NLS server can request the device to return multiple location reports during an active location MT-LR or NI-LR procedure by using a UBP Request or Provide Location Information exchange.

Dual-stack Support

The 5G LMF supports IPv4/IPv6 dual-stack communication with the 5G core nodes. To enable it, set the `enl-5g-proxy.service.externalIPFamilyPolicy` parameter to **`PreferDualStack`** when the 5G LMF Proxy is used. Otherwise, set the `nls-ts.lmf.service.externalIPFamilyPolicy` to **`PreferDualStack`**.

To check which deployment method will be used, see the *5G LMF Proxy Service* section in the Related Information.

For more details about the parameters, see the *NLS Configuration* section in the *Installation instruction*.

In case of multiple NRF settings, the primary, secondary, and tertiary URLs can be configured as IPv6 addresses.

In case of AMF, the 5G LMF can communicate on IPv6 during all types of positioning, for example, exchanging `DetermineLocation` messages. AMF cache works the same way as with IPv4.

When dual stack is configured, LMF will listen on any interface regardless of IP version.

—— RELATED INFORMATION ——

5G LMF Proxy Service

Positioning Control Data File and CLI Description

5G RDS Positioning Technique

Parameter List, NLS

NLS Configuration, Installation Instruction

2.2 LMF Indirect Communication

Operations

ENL supports indirect communication Model C option 1 and option 2, as described below.

Indirect Communication Model C Option 1

In the case of active indirect communication, the nodes of LMF and AMF are using the `3gpp-sbi-target-apiRoot` header in the request of the following operations:

— N1N2MessageTransfer (NLS->AMF) using the `Namf_Communication` service

— N1MessageNotify (AMF->NLS) using the `Namf_Communication` service

— N2InfoNotify (AMF->NLS) using the `Namf_Communication` service

— DetermineLocation (AMF->NLS) using the `Nlmf_Location` service

The LMF sends the Server header in an erroneous response for the operations above.

In erroneous responses to the LMF, the following cases are possible:

— the AMF or the SCP adds the `Server` header

— the SCP adds the `Via` header

Indirect Communication Model C Option 2

If model C option 2 is enabled, in addition to the scenarios of Model C option 1, the LMF includes the following headers in the requests for target NF producer reselection by the SCP:

— `3gpp-Sbi-Discovery-target-nf-set-id`, containing the target NF Set ID which corresponds to the target NF selected by the NLS

— `3gpp-Sbi-Discovery-target-nf-service-set-id`, containing the target NF Service Set ID which corresponds to the target NF selected by the NLS

In addition, LMF can parse the `3gpp-Sbi-Producer-Id` header if it is present in the service response (200, 4XX, 5XX) from the SCP. The content of `3gpp-Sbi-Producer-Id` is ignored by NLS.

Proxy Provisioning

The address of the SCP nodes available in the network are provisioned in the NLS. More SCP instances can be configured. An SCP instance is configured with its FQDN, which consists of the following parts:

— scheme

— authority

— API index

Example: https://www.scp.com/a/b/c.

Behavior Model

Dedicated administration state switch has been introduced to enable and disable the indirect communication approach (see table *PPLMF Parameters on Cluster Level (5G Indirect Communication* in *Parameter List* for NLS). NLS performs round-robin load share among the SCP instances. NLS maintains single TCP session towards every provisioned SCP instance.

Note: There is no fallback to direct communication if all SCPs fail.

NLS supports SCP monitoring functionality using HTTP/2 PING frame with configurable interval between consecutive pings.

The communication between NLS and AMF can also be asymmetric ("mixed communication cases"), where the parties separately either involve or leave out SCP from the message handling.

The communication between LMF and NRF remains direct.

Note: Some positioning session might require more than one LPP/NRPPa (N1/N2) request to the AMF. In this case, LMF uses the same SCP for the entire session, unless reselection due to an SCP failure is needed.

If Model C option 2 is enabled and NRF does not send `nfSetIdList` and `nfServiceSetIdList` during NF Discovery, LMF will fall back to Model C option 1.

2.2.1 Generic 3GPP Indirect Communication Flow

Note: Steps 1-3 and step 7 are done by the NF service consumer.

1. NF target selection based on cache or NF discovery
2. SCP selection from provisioned available SCPs in a round-robin manner
3. Request construction where the target URI is a combination of the SCP apiRoot and the service provider apiPath.
    - Additional headers:
        - `3gpp-sbi-target-apiRoot`: contains the apiRoot of the service provider
        - `Authorization`: included if Oauth2 is enabled
    - If the 5G Indirect Communication Model C Option 2 is enabled, the service consumer also adds the following additional headers to the request along with the headers above:
        - `3gpp-Sbi-Discovery-target-nf-set-id` contains the target NF Set ID.
        - `3gpp-Sbi-Discovery-target-nf-service-set-id` contains the target NF Service Set ID.
        - NF consumer adds this info from the cached `nfSetId` and `nfServiceSetId` received during NF producer discovery, according to the conditions below:
            - If the NRF does not send `nfSetIdList` and `nfServiceSetIdList` during NF Discovery, the service request

towards SCP will not include the `3gpp-Sbi-Discovery-*` headers.

- If any or both of `nfSetIdList` and `nfServiceSetIdList` are present along with `plmnList` configuration (NF Profile configuration parameter of HTTP2 component), and if no `setIds` match the PLMN ID configured in `plmnList`, a fallback will happen to Model C Option 1 and the service request will not include `3gpp-Sbi-Discovery-*` headers.
- If `plmnList` is empty, and if any or both of `nfSetIdList` and `nfServiceSetIdList` are present in the Discovery response, the first entry (`setid`) from the list will be selected.
- If there are multiple `setids` matching the same PLMN, the first matching set id is selected from `nfSetIdList` and from `nfServiceSetIdList`.

— The `3gpp-Sbi-Discovery-*` headers can be used by can be used by the SCP to reselect the target NF producer if the NF producer instance selected by the NF consumer cannot serve the request.

4. SCP forwarding request to service provider
   - Replacing SCP apiRoot with the content of `3gpp-sbi-target-apiRoot`
   - Removing `3gpp-sbi-target-apiRoot` header
5. Response construction in NF Producer
   - Addition of `Server` header in erroneous response
6. SCP forwarding response to service consumer
   - In case of NF producer reselection, the SCP sends the service response (`200`, 4XX, 5XX) to the NF consumer with the `3gpp-Sbi-Producer-Id` header, which contains any or all of `instanceId`, `serviceInstanceId`, `nfSetId`, and `nfServiceSetID` of the NF producer.
   - Addition of `Server` header in the created erroneous response

     Note: If the response received from the service producer already contains a `Server` header, the SCP copies and forwards that header.
   - Addition of `Via` header in the received erroneous response if:

     — the SCP received an erroneous response from the service producer and there was no `Server` header in it
7. Determining the source of error
   - Trying with new SCP if the source was the immediate SCP

At the current state, NLS does not close the serving interface if there is no available SCP and the indirect communication mode is configured. As a solution, NLS sends every 100th outgoing request to a randomly selected SCP entry from the provisioned SCP list. If the selected SCP successfully dispatches the messages, it is considered to be active again in the NLS administration.

2.2.2 Basic Indirect Operation Flow Handling

NLS Positive Flow

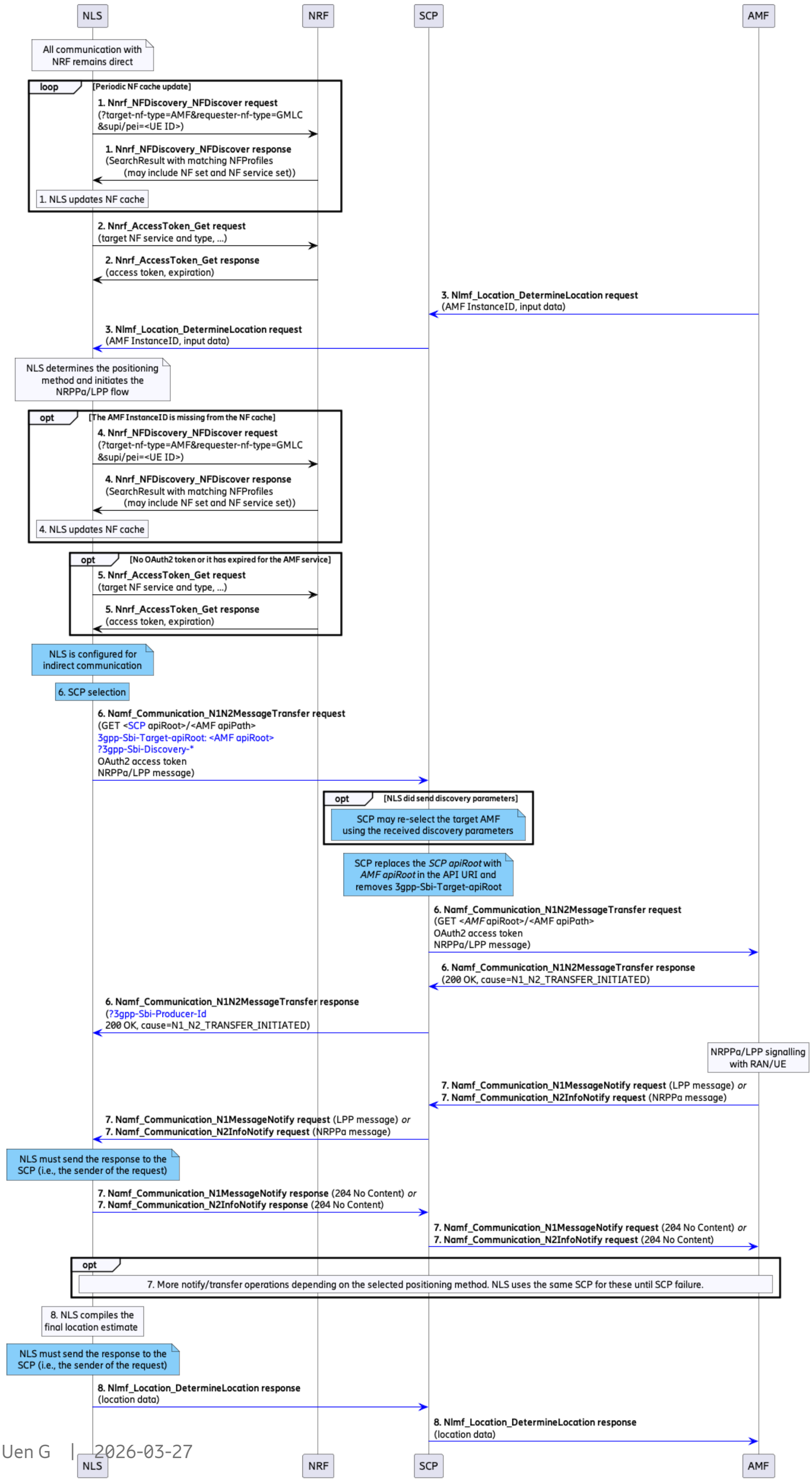

Figure 4 NLS Positive Flow

The flow graph above depicts the location service request procedure, in which the NLS is responsible for the following:

— Positioning method selection

— Positioning method procedure control

and where the incoming request is coming through the SCP node on behalf of AMF.

1. This step remains direct and the same as in the legacy direct positioning flow case.

2. This step remains direct and the same as in the legacy direct positioning flow case.

3. The AMF sends the `Nlmf_Location_DetermineLocation` request to the NLS via the proxy node called SCP.

   The `Nlmf_Location_DetermineLocation` request contains an LCS Correlation identifier, the serving cell identity of the target UE, and might contain the serving AMF instance ID.

   NLS determines the positioning method and initiates the NRPPa/LPP flow.

4. This step occurs if the AMF Instance ID is missing from the NF cache. In this case, NLS updates the cache with the latest information.

5. This step occurs if the OAuth access token is not in the cache, or has expired.

6. NLS is configured for indirect communication:

   — NLS selects the SCP from the provisioned ones in a round-robin manner.

   — Optional: In case Model C Option 2 is enabled and NLS is able to add the discovery parameters to the request, the SCP reselects the target AMF if the target AMF instance selected by the NLS cannot serve the request.

   — NLS inserts the API root of the request owner AMF in the header of 3gpp-Sbi-Target-apiRoot: *`<AMF API root>`*

   — SCP notifies the message and replaces the SCP apiRoot with the AMF apiRoot

   — The `Namf_Communication_N1N2MessageTransfer` request-response exchange is done in both LPP and NRPPa cases

   — Optional: In case of NF producer reselection, the SCP adds the `3gpp-Sbi-Producer-Id` header to the `Namf_Communication_N1N2MessageTransfer` response that it sends to the NLS, containing identifiers of the AMF instance selected by the SCP.

7. If the AMF is also using the SCP:

— `Namf_Communication_N1MessageNotify/` `Namf_Communication_N2InfoNotify` request-response exchange is done via the SCP node in both directions

— According to the normal operation, LMF sends HTTP "204 No Content" response code

— Optional: more notify/transfer operations depending on the selected positioning method. NLS uses the same SCP for these until SCP failure.

8. NLS compiles the final location estimate:

   - The Nlmf_Location_DetermineLocation response is sent to the SCP
   - SCP forwards the message

NLS Positive Flow Mixed Case, NLS is Configured for Direct Communication

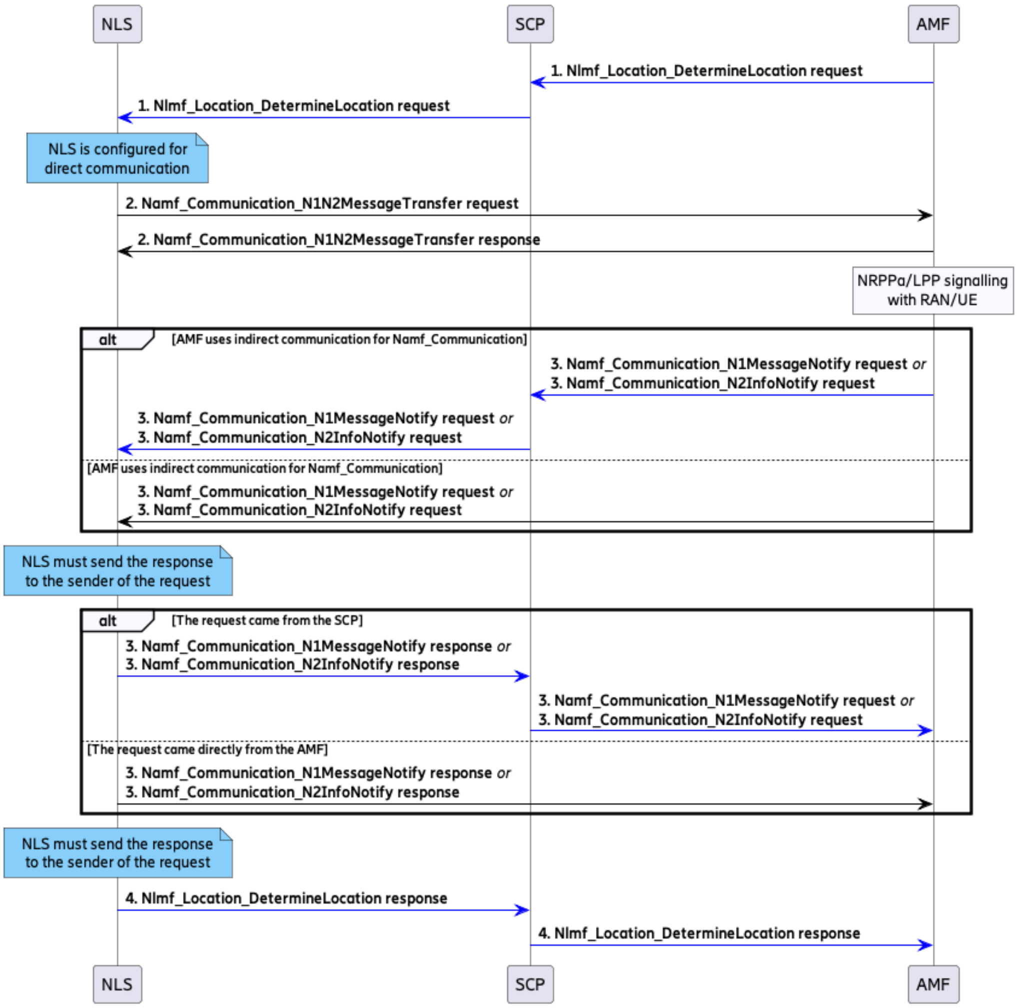

Figure 5 NLS Positive Flow Mixed Case, NLS is Configured for Direct Communication

- NLS sends indirect answer to indirect `Nlmf_Location_DetermineLocation` request.
- NLS (as a request initiator) sends the `Namf_Communication_N1N2MessageTransfer` requests directly to the AMF.
- Depending on the source of the incoming `Namf_Communication_N1MessageNotify/Namf_Communication_N2InfoNotify` request, the response is sent back to the same entity.

NLS Positive Flow Mixed Case, NLS is Configured for Indirect Communication

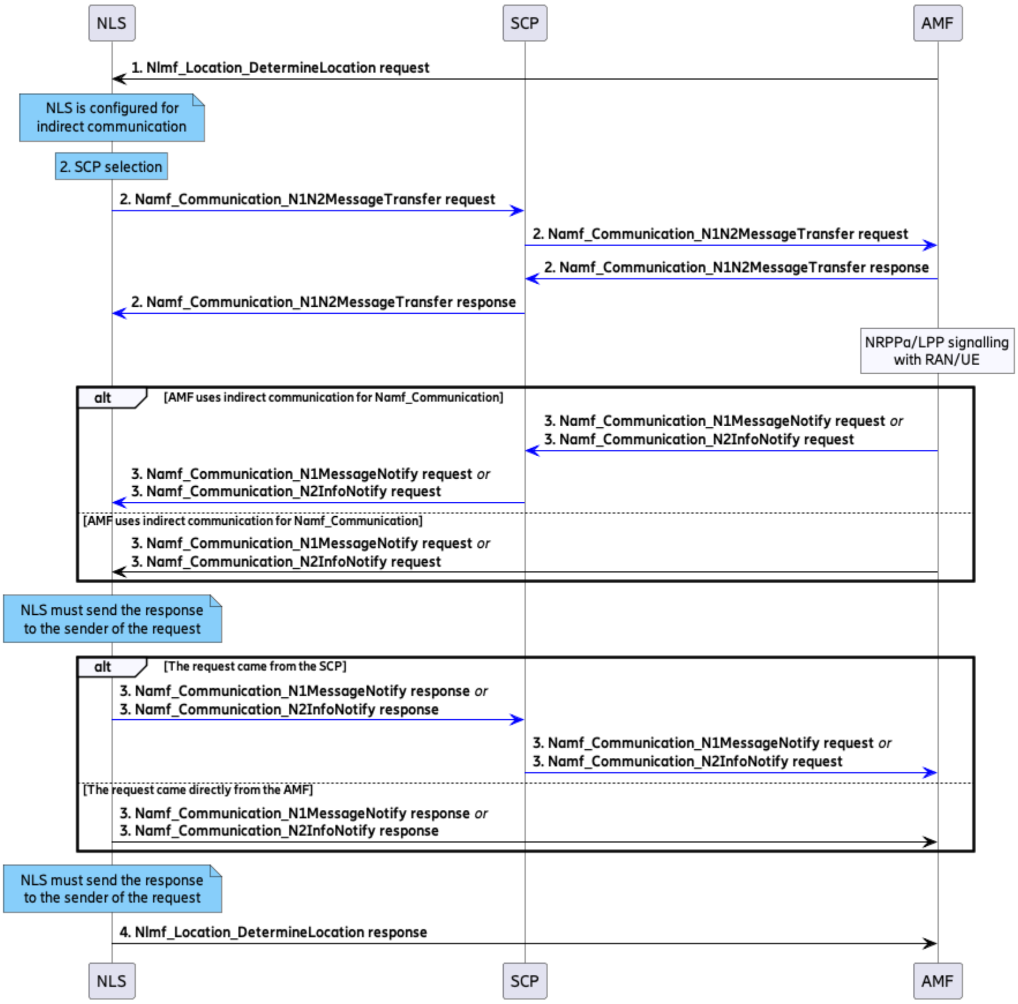

Figure 6 NLS Positive Flow Mixed Case, NLS is Configured for Indirect Communication

- NLS sends direct answer to direct `Nlmf_Location_DetermineLocation` request.
- NLS (as a request initiator) sends the `Namf_Communication_N1N2MessageTransfer` requests towards the selected SCP entity.
- Depending on the source of the incoming `Namf_Communication_N1MessageNotify/ Namf_Communication_N2InfoNotify` request, the response is sent back to the same entity.

2.2.3 Error Handling during Indirect Communication

LMF Error Handling

The error handling described in this subsection is applicable to the following service operations initiated by the AMF:

- Nlmf_Location_DetermineLocation
- Namf_Communication_N1MessageNotify
- Namf_Communication_N2InfoNotify

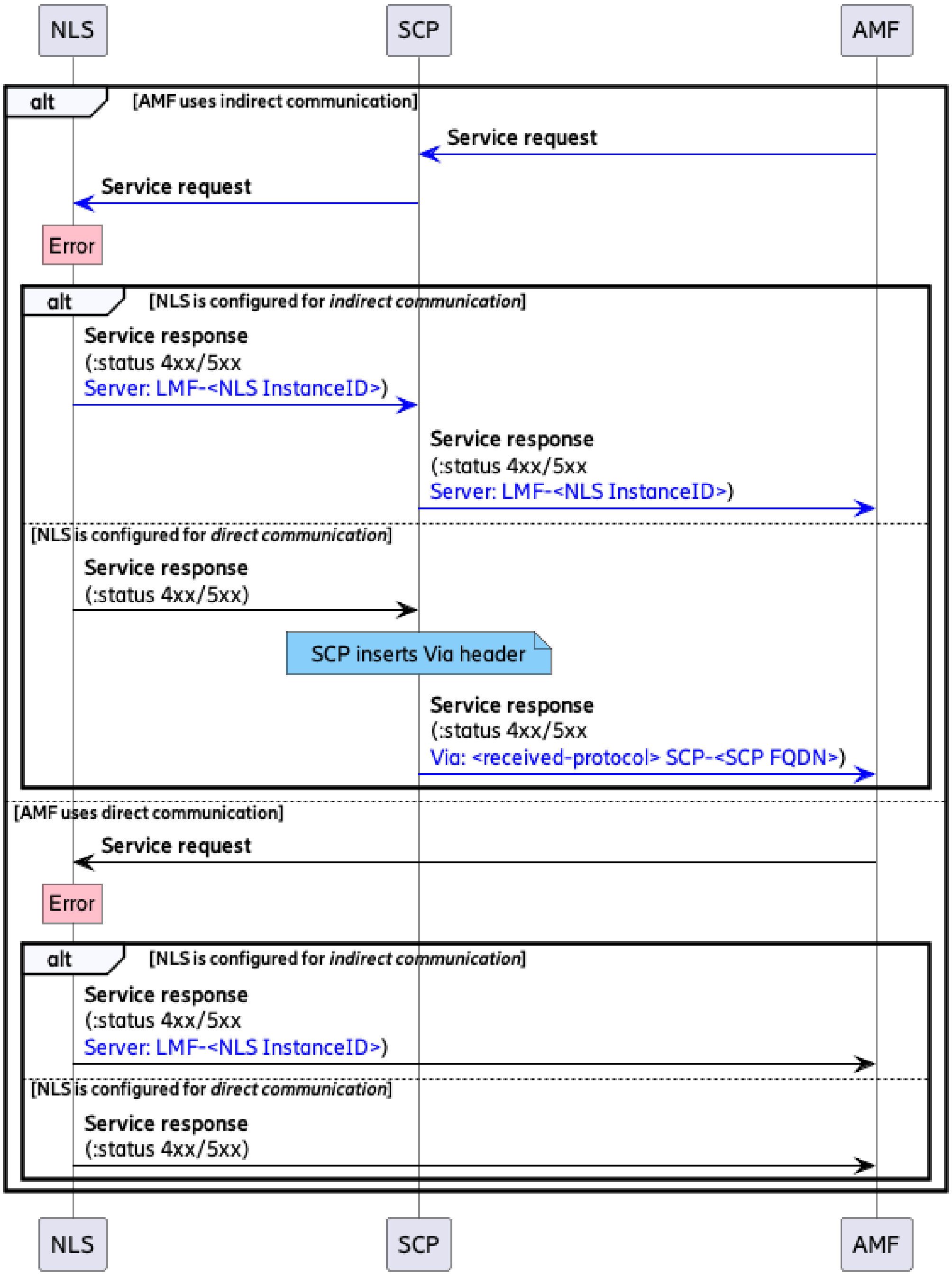

Figure 7 NLS Negative Flow - Error Handling

Depending on the AMF communication strategy, AMF uses direct or indirect communication.

In the case of indirect AMF communication:

— Indirect NLS sends a response stating the problematic LMF instance ID in the `Server` header to the same SCP instance.

— Direct NLS sends a response to the SCP without the `Server` header, and SCP adds a `Via` header with the SCP FQDN.

In case of direct AMF communication:

— Indirect NLS sends a response stating the problematic NLS instance entity in the `Server` header directly to the requestor AMF.

— Direct NLS simply sends a 4xx/5xx response without the `Server` header, directly to the requestor AMF .

Producer Error Handling

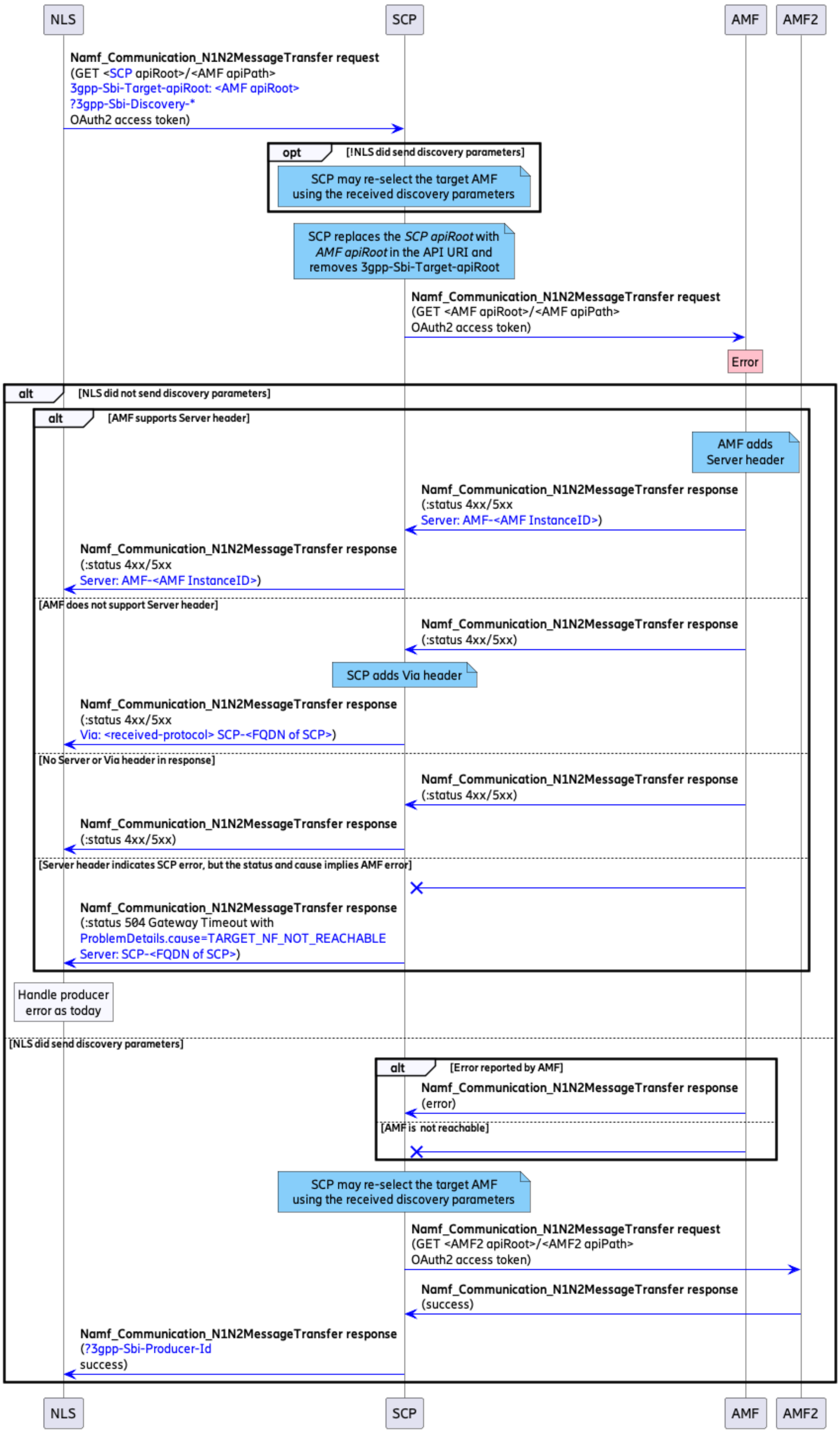

Figure 8 NLS Negative Flow - Producer Error

NLS as a consumer can get notified of producer errors in multiple ways:

— AMF as a producer adds a `Server` header in the 4xx/5xx responses.

— AMF skips adding the `Server` header, SCP in this case adds a `Via` header to the response, including its own FQDN.

— Neither AMF, nor SCP populate the header set of the 4xx/5xx response.

— SCP returns an 504 Gateway Timeout specifying TARGET_NF_NOT_REACHABLE as cause and the FQDN of the SCP in the `Server` header.

Note: In case Model C option 2 is enabled and the SCP is able to reselect the target AMF, the NLS is not notified of the original producer error.

SCP Error during Communication

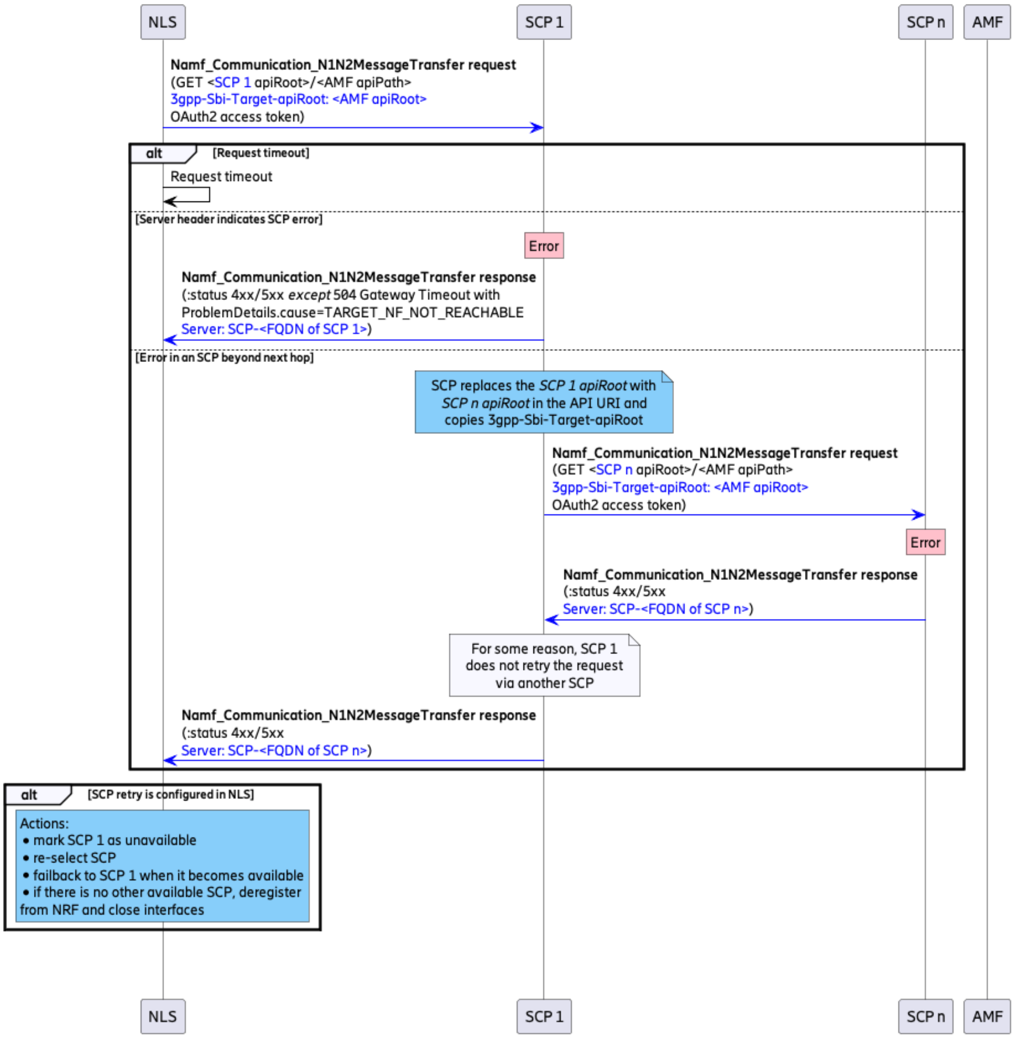

Figure 9 NLS Negative Flow - SCP Error

NLS is capable of handling the following scenarios:

- No response from the serving SCP
- SCP Namf_Communication_N1N2MessageTransfer response with a 4xx/5xx (with the exception of 504, cause= TARGET_NF_NOT_REACHABLE) status with the problematic SCP FQDN
- The chosen SCP does not retry forwarding the request to any other SCP and sends back a 4xx/5xx Namf_Communication_N1N2MessageTransfer response with a `Server` header containing the problematic SCP FQDN.

2.3 Signaling Procedures

This section describes the signaling procedures supported by the NLS LMF value package.

2.3.1 Procedures for 5G Location Services

Figure 10 illustrates the general positioning procedure in the LMF. This procedure is applicable to both emergency positioning (for example, 5GC-NI-LR) and to normal requests from an LCS client (for example, 5GC-MT-LR).

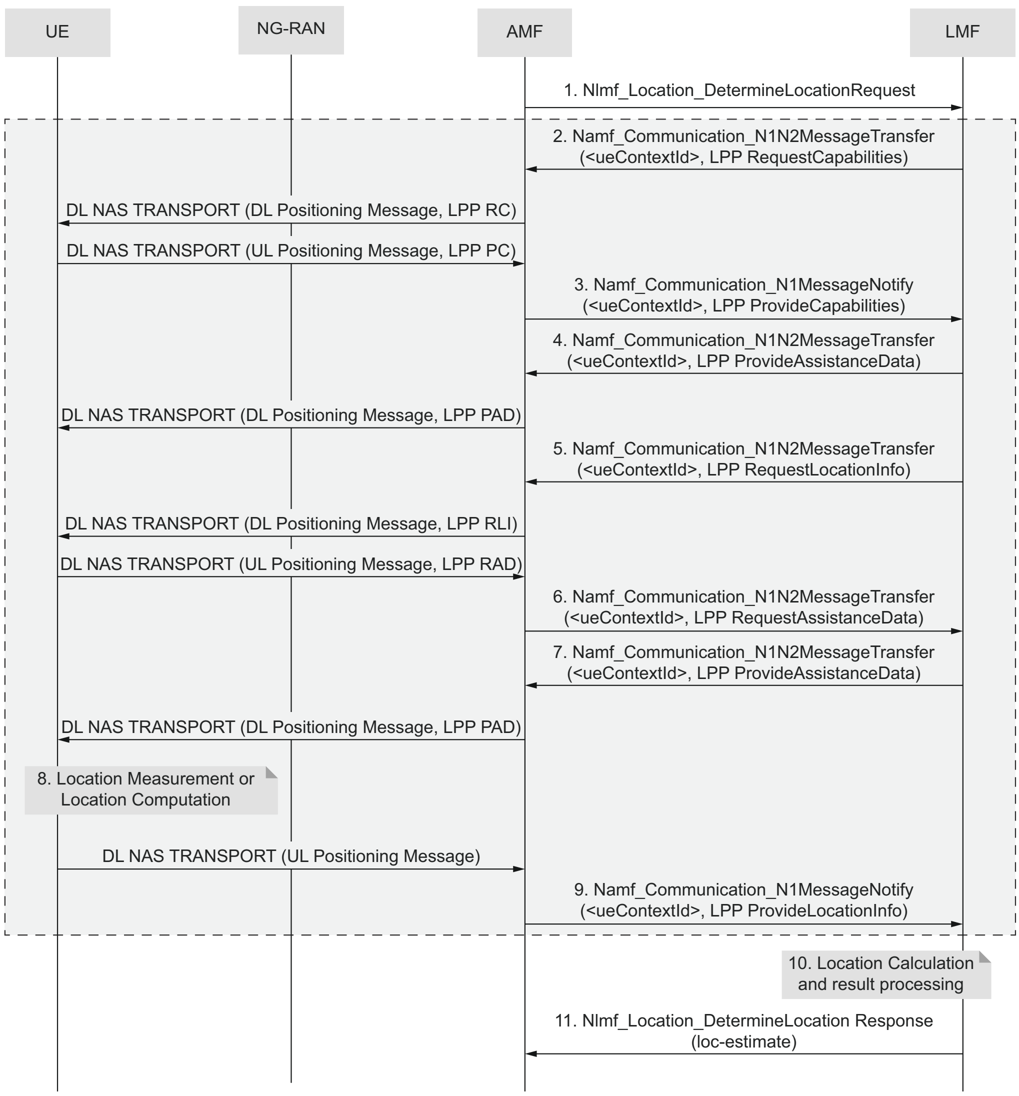

Figure 10 General Positioning Procedure in LMF through LPP

Steps

1. The AMF selects an LMF instance based on an NRF query or the configuration, and invokes the `Nlmf_Location_DetermineLocation` service operation toward the LMF to request the current location of the UE. The service operation includes an LCS correlation identifier, the serving NR CID, and the client type. The required QoS and the supported GAD shapes can also be included. If OAuth 2.0 is configured, a valid access token must be provided in the request header.

The AMF identity must be included if Step 2 to Step 8 are used.

2. If any LPP-based positioning method (for example, A-GNSS) is enabled, the LMF sends the LPP `RequestCapabilities` message to the UE.

   The LMF invokes the `Namf_Communication_N1N2MessageTransfer` service operation toward the AMF to transfer a `Downlink Positioning Message` to the UE, and includes the LPP binary data as the N1 message content. The AMF forwards the `Downlink Positioning Message` to the UE in a `DOWNLINK NAS TRANSPORT` message.

3. The UE returns the LPP `ProvideCapabilities` message to the LMF with UE positioning capabilities.

   The UE sends back an `Uplink Positioning Message` to the AMF included in a `NAS Transport` message. Then, the AMF invokes the `Namf_Communication_N1MessageNotify` service operation toward the LMF, including the `Uplink Positioning Message` received from the UE and the LCS correlation identifier.

For all LPP messages exchanged between the LMF and the UE, the AMF repeats Step 2 and Step 3.

4. The LMF selects the positioning method based on the UE positioning capabilities, the positioning method configurations, and the parameters included in the `Location Request` message.

   If the LMF selects A-GNSS, it generates the default A-GNSS assistance data based on the reference location derived from the serving NR CID. The LMF then sends the A-GNSS assistance data to the target UE in an LPP `ProvideAssistanceData` message.

5. The LMF sends an LPP `RequestLocationInformation` message to the UE to get the A-GNSS measurement data (UE-assisted) or the location estimate (UE- based or autonomous).

6. The UE might request more A-GNSS assistance data from the LMF by sending an LPP `RequestAssistanceData` message.

7. The LMF delivers the A-GNSS assistance data requested by the UE in an LPP `ProvideAssistanceData` message.

Step 6 and Step 7 can be repeated if more A-GNSS assistance data is needed.

8. The UE stores the A-GNSS assistance data and performs any positioning measurements or location computation requested by the LMF.

9. The UE returns an LPP `ProvideLocationInformation` message to the LMF, which includes either A-GNSS measurements (UE-assisted) or a location estimate (UE-based).

10. The LMF processes the location information returned from the UE. For the UE-assisted A-GNSS method, the LMF calculates a position estimate from the measurements.

    Based on the positioning method configuration, the LMF might use other positioning methods too and select the appropriate location result from the available position estimates.

    AGNSS is the first performed method with UBP over LPPe.

11. The LMF returns the `Nlmf_Location_DetermineLocation` service operation to the AMF, including the current location of the target UE. The service operation includes the location estimate, together with its age and accuracy. The service operation can also include information about the positioning method.

    Note: For LPP messages, the LMF can both request acknowledgment from the UE and respond to the acknowledgment if required by the UE. When sending an LPP request to the UE, the LMF requests acknowledgment from the UE. When receiving an LPP request from the UE, the LMF returns an LPP acknowledgment message if acknowledgment is required by the UE.

    According to *3GPP 29.572*, the location data is given in meter and degree. Value of some data types is float. For the location result, the LMF by default keeps 9 decimals on floats and 17 on doubles. The default value is configurable through the `LocationDataMaximumDecimals` parameter. By default, in `VelocityEstimate`, the LMF keeps one place of decimal on `hUncertainty` and `vUncertainty`. In `LocationData`, the LMF keeps two places of decimals on altitude.

2.3.2 NF Service Framework Procedures

Figure 11 illustrates how LMF supports the Network Function Service framework procedure by interacting with the NRF.

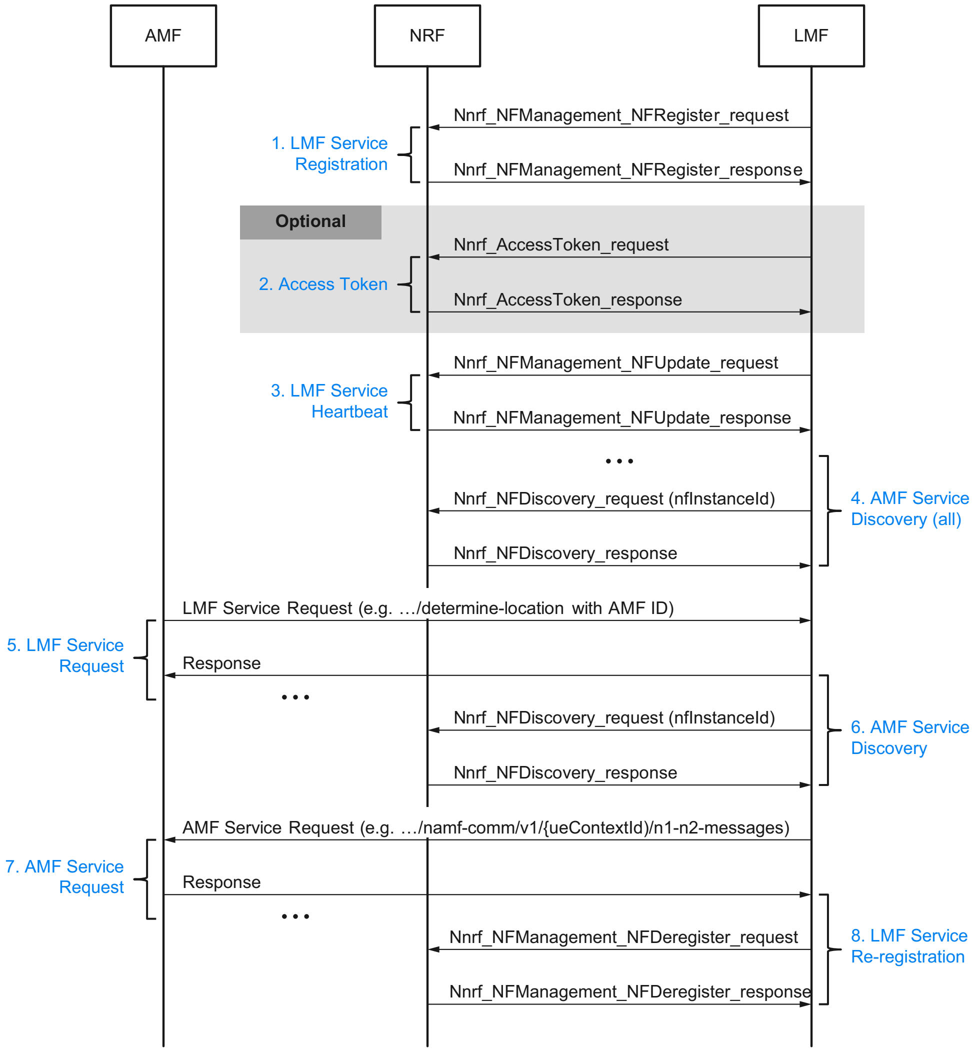

Figure 11 NF Service Framework Procedure

Steps

1. The LMF registers the service.

   a. The LMF sends an `Nnrf_NFManagement_NFRegister Request` message to the NRF with its NF profile included when the LMF becomes operational for the first time, or in case of a failover.

      Note: The LMF NF profile is generated based on the local NF profile template and configuration parameters. For more details, see the Related Information.

   b. The NRF stores the NF profile of the LMF and marks the NF service instance available.

The NRF might return a heartbeat timer containing the number of seconds expected between two consecutive heartbeat messages from an NF instance to the NRF.

2. The LMF NF profile is generated based on the local NF profile(Optional) The LMF requests an access token for the NRF and the AMF services. The access token is provided by the NRF with additional information, such as the expiration time or the scope. This step can be configured by enabling OAuth 2.0.

3. The LMF provides the service heartbeat information.

    a. The LMF provides the heartbeat of the NF service instance by periodically sending `Nnrf_NFManagement_NFUpdate Request` messages to the NRF .

    b. The NRF acknowledges the `Nnrf_NFManagement_NFUpdate Request` message, and accepts it by sending an `Nnrf_NFManagement_NFUpdate Response` message.

4. The LMF sends a discovery request and caches the received addresses.

    a. When the LMF starts up, it queries the NRF for all available AMF service instances in the network. The LMF invokes the `Nnrf_NFDiscovery_Request` message from the NRF with the following values:

        — `AMF` for `target-nf-type`

        — `namf-comm` for `service-names`

    b. The NRF determines a set of NF instances matching the `Nnrf_NFDiscovery_Request` message, and sends the NF profiles of the determined NF instances in an `Nnrf_NFDiscovery_Response` message. Each NF profile contains the address of the service instance for sending N1/LPP and N2/NRPPa messages, for example, `ipEndPoints`.

    c. The LMF caches the addresses of the AMF service instances based on the discovery result returned by the NRF.

Note:

— The validity time of each cache item is set to the minimum value between the `validityPeriod` value returned from the NRF and the `NRF Client Recycle Interval` parameter. Before the validity time expires, the LMF queries the NRF for all AMF instances and updates the cache accordingly. The timing of the query is the minimum of `NRF Default Periodic Discovery Interval` and `validityPeriod` minus `NRF Processing Time`. The cache is not removed in case a new service discovery to the NRF fails.

— This kind of discovery request is disabled if the `AMF Subscription Switch` is set to `individual`. In this case, only the AMF positioning procedure initiates the `Nnrf_NFDiscovery` operation.

5. The AMF sends a location service request to the LMF.

    a. The AMF finds the LMF service instance through an NRF query or the local configuration.

    b. The AMF sends an `Nlmf_Location_DetermineLocation Request` message to an LMF instance. This message contains the `amfId` parameter, which is the NF ID of the LMF instance.

6. The AMF initiates an `Nnrf_NFDiscovery` operation during the positioning procedure.

    a. The LMF needs to get the AMF address to send an LPP message to the target UE. The LMF first tries to find the AMF address in the local cache using the `amfId`. If it does not find it there or if the validity of the cache item is expired, the LMF invokes an `Nnrf_NFDiscovery_Request` message with the AMF instance ID received from the `Location Request` message.

    b. The NRF sends the NF profile of the specified NF instance in an `Nnrf_NFDiscovery_Response` message. The LMF gets the AMF service address from the NF profile and updates the cache.

    Note: If the AMF discovery fails or it is not possible, then the LMF can take the source IP address from the `Nlmf_Location_DetermineLocation` request and the port value defined in the `AMF_RestPort` configuration parameter to send the subsequent LPP/NRPPa requests to. For more details about the parameter, see the Related Information.

7. The LMF sends LPP requests to the UE through the AMF.

    a. For the `Downlink Positioning Message`, the LMF sends LPP requests to the UE through the AMF `Namf_Communication_N1N2MessageTransfer` service operation using the AMF address obtained in Step 6.

   b. For the `Uplink Positioning Message`, the AMF gets the LPP message from the UE and forwards it to the callback address of the LMF instance through the `Namf_Communication_N1MessageNotify` service operation.

8. The LMF deregisters the service.

   a. When the LMF becomes unavailable, for example, when it is about to gracefully shut down or disconnect from the network, the LMF sends an `Nnrf_NFManagement_NFDeregister_Request` message to the NRF to inform it of its unavailability.

   b. The NRF marks the LMF service instance unavailable. The NRF acknowledges the `Nnrf_NF Deregistration` procedure and accepts it in an `Nnrf_NFManagement_NFDeregister_Response` message.

—— RELATED INFORMATION ——

5G LMF Proxy Service

Parameter List, NLS

2.3.2.1 NfType-Based AMF Monitoring

AMF monitoring can be configured as `global` to trigger an alarm when a known AMF status changes to `suspended`. The `global` configuration means that the LMF creates an `NfType`-based subscription. This setting is proposed if the NRFs are aware of all AMFs so they can notify the user of all changes. In this case, after the first successful `NfType`-based discovery response, if the **NRF Subscription Switch** is set to `global`, the LMF starts monitoring the AMF statuses by subscribing to the NRF for AMF status changes. This means that the LMF invokes an `Nnrf_NFManagement_NFStatusSubscribe_Request` message where the `NfType` is AMF.

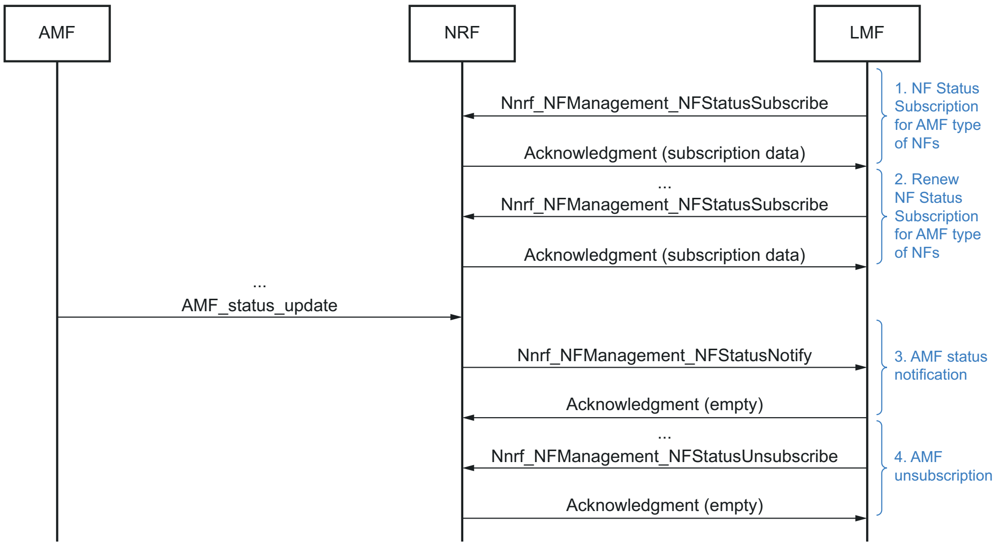

Figure 12 NfType-Based AMF Monitoring

Steps

1. The LMF creates an NF Status Subscription.

    a. The LMF sends the `Nnrf_NFManagement_NFStatusSubscribe` request, which includes the specification to be notified about in case of changes regarding the AMF instances.

    b. The NRF replies with either an `Acknowledgment` and all the subscription-related data including the validity time, or an error message upon a failed subscription.

    With the NF Status Subscription, the LMF can be notified by the NRF when AMF instances are registered or deregistered in the NRF, or when an AMF instance profile is modified.

    Note: The subscription can be created on a different NRF, and the Location Header of the `Acknowledgment` contains the URL of the subscription created on this NRF. In this case, the rest of the communication takes place between the LMF and the NRF where the subscription is created.

2. The LMF renews the NF Status Subscription on the NRF before it expires.

    a. The LMF sends the `Nnrf_NFManagement_NFStatusSubscribe` message to the NRF with an updated validity time.

    b. The NRF replies with either an `Acknowledgment` and all the subscription-related data including the validity time, or an error message upon a failed subscription update. In case an error message is returned, the LMF attempts the subscription once more, then, after a second failure, it tries to subscribe to the NRF again. If the third attempt results in an error message too, the subscription update fails.

3. The NRF notifies the LMF for the AMF status change.

    a. If the status of an AMF changes, it sends an `AMF_status_update` message to the NRF.

    b. The NRF notifies the LMFs that subscribed for the status changes of the given AMF by sending them a `Nnrf_NFManagement_NFStatusNotify` message.

    c. The LMF replies with an `Acknowledgment`.

4. The LMF unsubscribes from the AMF status changes.

    a. The LMF sends a `Nnrf_NFManagement_NFStatusUnsubscribe` message to the NRF.

    b. The NRF replies with an `Acknowledgment`.

2.3.2.2 NfInstanceId-Based AMF Monitoring

AMF monitoring can be configured as `individual` to trigger an alarm when a known AMF status changes to `suspended`. The `individual` configuration means that the LMF creates an `NfInstanceID`-based subscription for all known AMFs on the active NRF. This setting is proposed in case of a hierarchical NRF setup where an NRF is not aware of all AMFs, so subscriptions are to be created on the NRF which is aware of the interested AMF.
In case the AMF starts a positioning session when the **NRF Subscription Switch** is set to `individual`, the LMF starts monitoring the AMF status by subscribing for AMF status changes to the NRF. It means that the LMF invokes an `Nnrf_NFManagement_NFStatusSubscribe_Request` message where the `NfInstanceId` of the current AMF is given as the `NfInstanceId`.

When the LMF component is gracefully deactivated, it stops monitoring the AMF statuses, and unsubscribes from the NRFs.

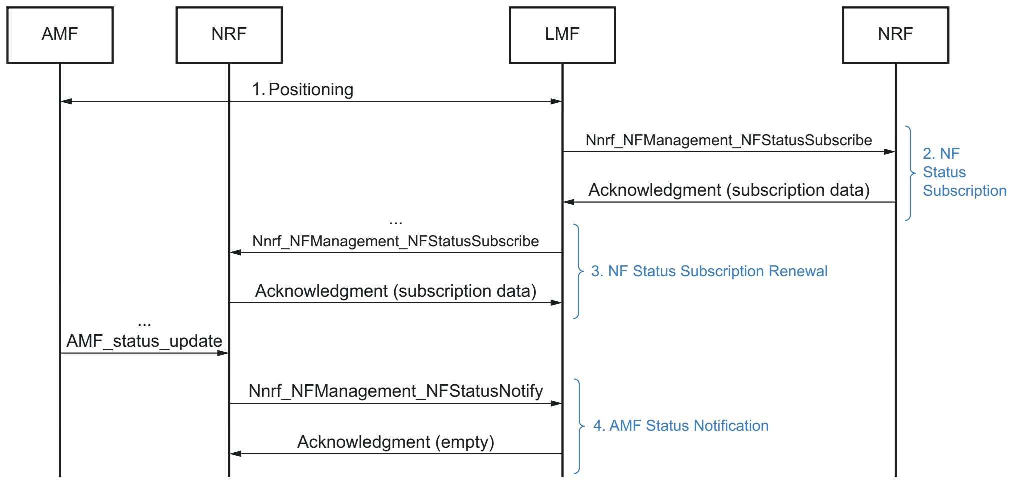

Figure 13 NfInstanceId-Based AMF Monitoring

Steps

1. The AMF conducts a positioning procedure with the LMF.

2. The LMF creates an NF Status Subscription.

    a. The LMF sends the `Nnrf_NFManagement_NFStatusSubscribe` request, which includes the specification to be notified about in case of changes regarding that AMF instance.

    b. The NRF replies with either an `Acknowledgment` and all the subscription-related data including the validity time, or an error message upon a failed subscription. If a subscription is created on another NRF, then the address of the created resource is present in the Location Header.

        Note: It is possible to create the subscription on a different NRF and the Location Header of the `Acknowledgment` contains the URL of the subscription created on that NRF. In this case, the rest of the communication takes place between the LMF and the NRF where the subscription is created.

    c. In case of a failed subscription, the subscription is attempted again on the same NRF at first, then on the other configured NRFs.

3. The LMF renews the NF Status Subscription on the NRF before it expires.

    a. The LMF selects the earliest expiration time. When it is passed, the LMF sends an `Nnrf_NFManagement_NFStatusSubscribe` message to the NRF with an updated validity time.

    b. The NRF replies with either an `Acknowledgment` and all the subscription-related data including the validity time, or an error message upon a failed subscription update. In case an error message is returned, the LMF attempts the subscription once more, then after a second failure, it tries to subscribe to the NRF again. If the third attempt results in an error message, then the subscription update fails.

4. The NRF notifies the LMF of all changes regarding the AMFs. The changes and how the LMF handles them are the same as in case of `NfType`-based AMF subscriptions.

   Note: In case of an `NfInstanceId`-based AMF monitoring, the LMF does not remove subscriptions.

2.3.3 UBP 5G Signaling Procedures

The following positioning methods are supported for UBP over LPPe:

— UE-A

— UE-B

— UE-A and UE-B

— Standalone

2.3.3.1 A-GNSS with UBP over LPPe

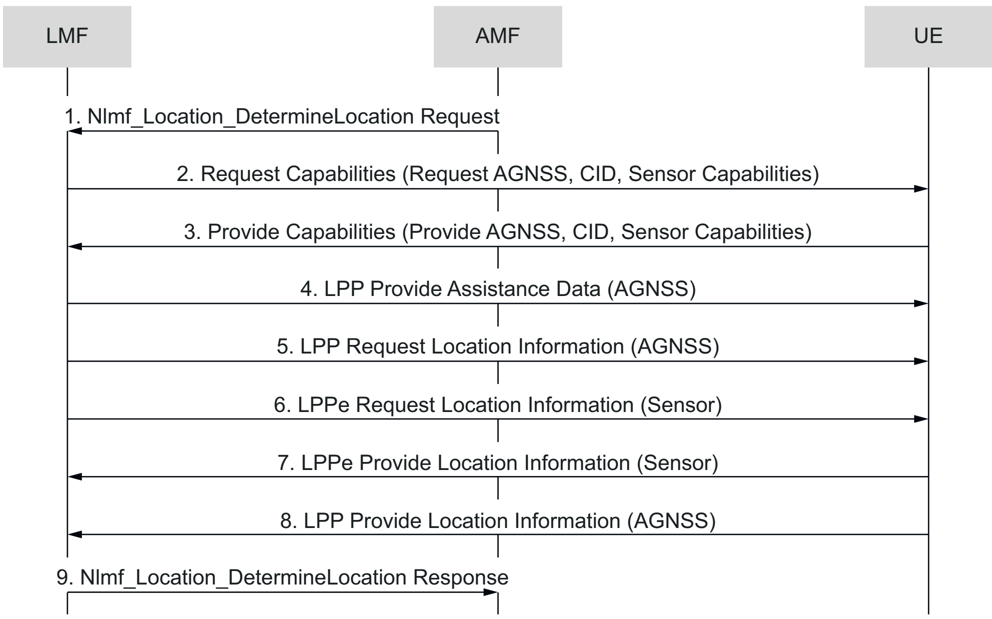

Figure 14 A-GNSS with UBP over LPPe

Steps

1. The LMF receives the `Nlmf_Location_DetermineLocation Request` from the AMF.
2. The LMF sends the `Request Capabilities` to the UE through the AMF.
3. The UE returns the `Provide Capabilities` to the LMF through the AMF. The LMF decides the positioning methods. In this case the A-GNSS is selected with UBP over LPPe.
4. The LMF sends the `LPP Provide Assistance Data` (for A-GNSS) to the UE.
5. The LMF sends the `LPP Request Location Information` (for A-GNSS) to the UE.
6. The LMF sends the `LPPe Request Location Information` (for Sensor) to the UE.
7. The UE returns the `LPPe Provide Location Information` (for Sensor).
8. The UE returns the `LPP Provide Location Information` (for A-GNSS).
9. The LMF sends the `Nlmf_Location_DetermineLocation Response` to the AMF.

2.3.3.2 A-GNSS with Multiple UBP over LPPe

The A-GNSS with Multiple UBP over LPPe signaling procedure is supported in the following cases:

— Multiple location is enabled

— The UE is capable of UBP

— The first LPPe UBP answer arrives before the LPP A-GNSS

— The first LPPe UBP answer contains DBH

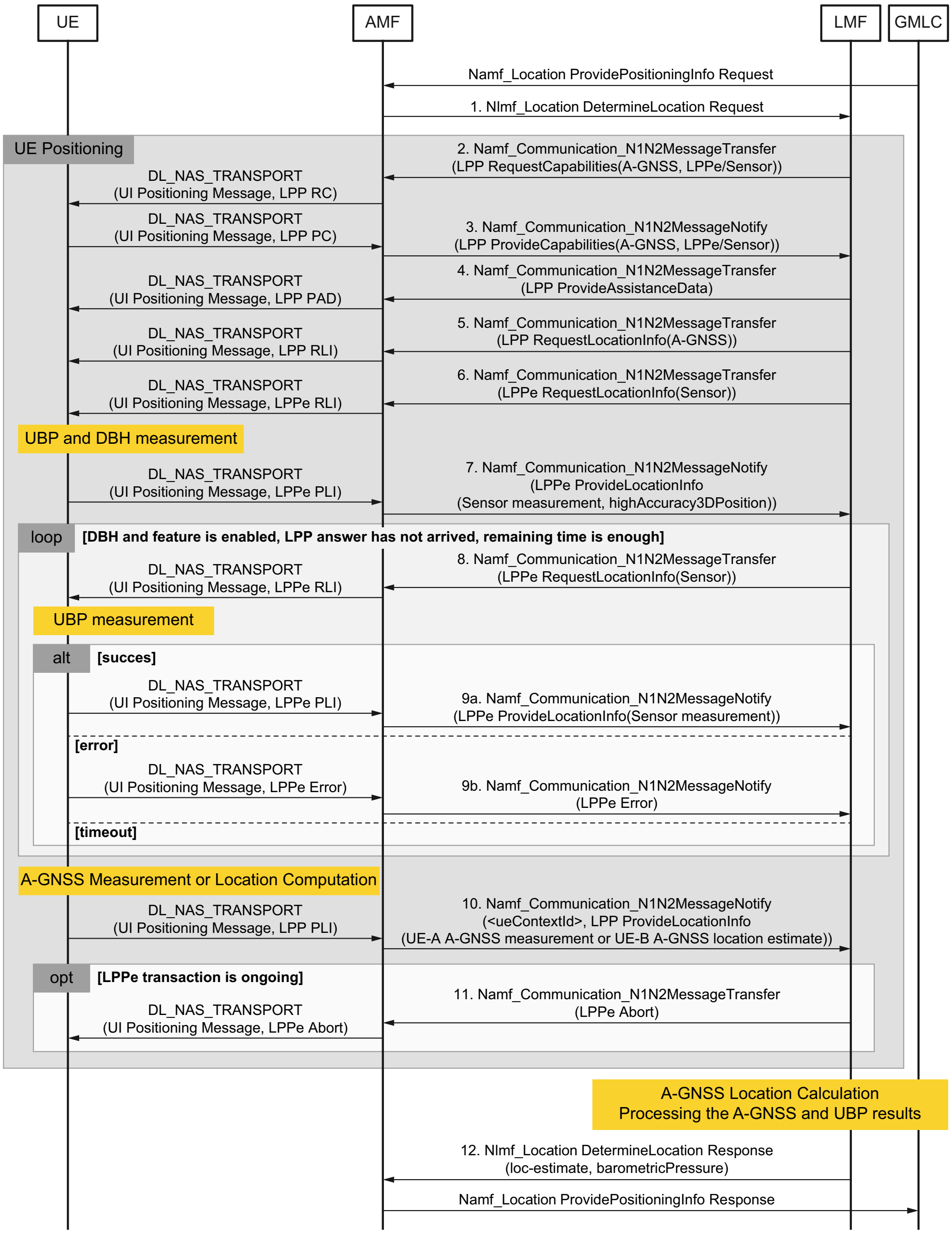

Figure 15 A-GNSS with Multiple UBP over LPPe

Steps

1. The AMF sends the `Nlmf_Location_DetermineLocation request` message to the LMF.

2. The LMF requests capabilities (LPP and LPPe) from the AMF.

   For the LPP, the message contains information related to A-GNSS, DBH or Civic Location (CL).

For the LPPe, the message contains sensor-related information.

3. The AMF provides capabilities to the LMF (the UE capable of A-GNSS and the sensor).

4. The LMF sends the `LPP ProvideAssistanceData` message to the AMF (A-GNSS AD).

5. The LMF sends the `LPP RequestLocationInfo` message to the AMF (A-GNSS/CL).

6. The LMF requests only the LPPe sensor information by sending the `LPPe RequestLocationInfo` message.

   The LPP response time depends on whether the procedure is executed for the first time or it is a repeated procedure.

7. The AMF responds with an `LPP/LPPe ProvideLocationInfo` message carrying an External Protocol Data Unit (EPDU) with sensor and optional DBH information.

8. The LMF sends an `LPPe RequestLocationInfo` message to request only LPPe sensor information.

   The LPP response time depends on whether the procedure is executed for the first time or it is a repeated procedure.

9. The AMF responds with an `LPP/LPPe ProvideLocationInfo` message carrying an EPDU with Sensor and optional DBH information.

   If the LPP answer is not arrived and there are enough time, repeat Step 8 and Step 9 (even in the case of error or timeout).

10. The AMF sends an `LPP ProvideLocationInfo` message carrying GNSS and EPDU with DBH information.

11. The LMF aborts any outstanding LPPe procedures for UBP.

    The NLS requests acknowledgement of the LPP message, but does not delay the completion of later steps if no acknowledgement is received from the AMF.

12. The LMF sends the `Nlmf_Location_DetermineLocation response` message to the AMF.

2.3.4 Altitude Selection

Altitude selection combines the DBH and the UE-A or UE-B AGNSS location estimates, based on vertical uncertainty. If location estimates are available from both methods, the location estimate with better horizontal accuracy is returned. However, if the location estimate with worse horizontal accuracy positioning method provides an altitude with lower vertical uncertainty, then that value is returned with the horizontal location estimate. The source of the altitude is noted in the positioning record. For information on the positioning record, see the Related Information.

For information on how to activate or deactivate altitude selection, see the *PPLMF Parameters on Cluster Level (A-GNSS Positioning)* table in the section mentioned in the Related Information.

Note: Altitude selection is used only if the positioning method preference parameter values are set to `Best Uncertainty`. For the parameters, see the *DBH Parameters for LPPe on LMF (Positioning Procedure)* table in the section mentioned in the Related Information.

For information about altitude selection-related counters, see the *LMF - General (Overall) - (Position Method)* table in the section mentioned in the Related Information.

—— RELATED INFORMATION ——

PositioningMethodAndUsage, Positioning Record File Description, NLS

1_190 59-HSC 901 1780

DBH for LMF, Parameter List, NLS

General (Overall), Statistics Description, NLS

2.3.5 Civic Address over LPPe Signaling Procedure

The Civic Address over LPPe signaling procedure is supported over LPP and LPPe as follows:

- LPPe Request Capabilities for DBH is supported in the LPP Request Capabilities message for GNSS.
- LPPe Provide Capabilities for DBH and Civic Address is supported in the LPP Provide Capabilities message for GNSS.
- LPPe Request Location Information for Civic Address is supported in the LPP Request Location Information message for GNSS.
- LPPe Provide Location Information for DBH and Civic Address is supported in the LPP Provide Location Information message for GNSS.

For more details, see the Related Information.

—— RELATED INFORMATION ——

Signaling Procedure of Civic Address over LPPe in the NR Network, DBH and Civic Address Positioning Technique

2.3.6 NR ECID Positioning Procedure

NR ECID is supported over NRPPa in the LMF as follows:

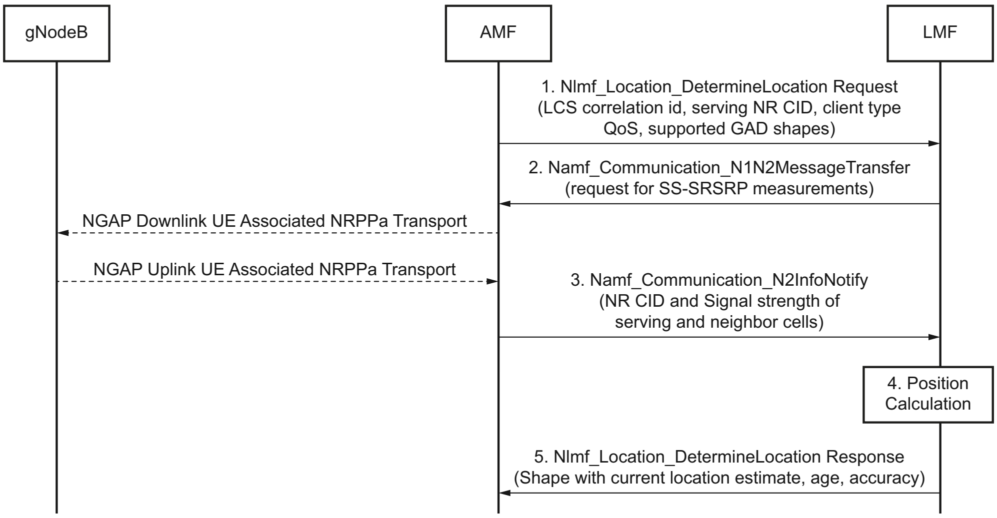

Figure 16 NR E-CID Message Flow

The NR ECID also gathers the Timing Advance Values.

The NR AECID functions similarly to the NR ECID, but gathers the Angle of Arrival values besides other values.

Steps

1. The AMF selects an LMF instance based on an NRF query or the configuration. The AMF invokes the Nlmf_Location_DetermineLocation service operation toward the LMF to request the current location of the UE. The service operation includes an LCS correlation identifier, the serving NR CID, and the client type. The required QoS and the supported GAD shapes can also be included.

2. The LMF selects the positioning method based on the positioning method configurations and the parameters included in the Location Request message. If the LMF selects an NR ECID positioning method, the LMF sends an NRPPa Measurement Initiation Request message to the gNodeB through the AMF to gather SS-RSRP measurement data from the UE.

3. The AMF returns the NRPPa Measurement Initiation Response message from the gNodeB to the LMF, which includes the signal strength values of the serving and the neighbor cells.

4. The LMF processes the location information returned from the AMF. For the NR ECID, the LMF calculates a position estimate from the measurements. Based on the positioning method configuration, the LMF might use other positioning methods too and select the appropriate location result from the available position estimates.

5. The LMF returns the Nlmf_Location_DetermineLocation service operation to the AMF, including the current location of the target UE. The service operation includes the location estimate together with its age and accuracy. The service operation can also include information about the positioning method.

2.4 Interfaces

The SBIs in the 5GC network use HTTP/2 with JSON as the application layer serialization protocol. The LMF uses the SBIs listed below for control plane positioning.

LMF Services

The LMF supports the NF service producer for the `Nlmf_Location` service per *TS 29.572*.

Table 1 SBI for LMF Services

Service Name	Service Operations	Operation Semantics
Nlmf_ Location	DetermineLocation	Request/Response

AMF Services

The LMF implements the NF service consumer for the `Namf_Communication` service per *TS 29.518*.

Table 2 SBI for AMF Services

Service Name	Service Operations	Operation Semantics
Namf_Communication	N1MessageNotify	Subscribe/Notify
	N1N2MessageTransfer	Request/Response
	N2InfoNotify	Subscribe/Notify

The LPP message is transported between the LMF and the UE using the `Namf_Communication` service provided by the AMF. HTTP multipart messages are used for N1/N2 message transfer, which includes one JSON body part and one or more binary bodies. The LCS N1 message encodes a 5GS NAS message for the LPP type, using the `vnd.3gpp.5gnas` content type.

Figure 17 shows the protocol stack for LPP transport in the 5GC network.

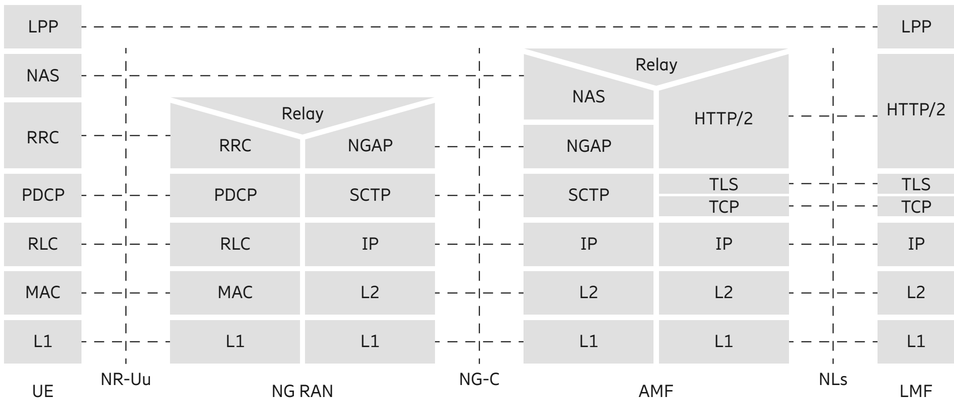

Figure 17 Protocol Stack for LPP Transport

Figure 18 shows the protocol stack for NRPPa transport in the 5GC network.

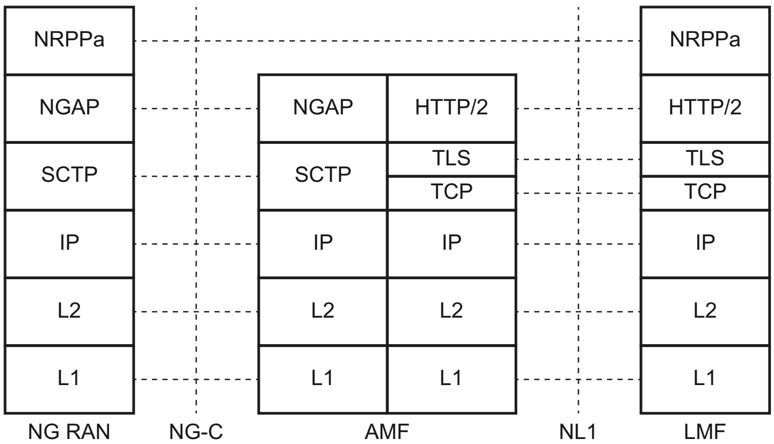

Figure 18 Protocol Stack for NRPPa Transport

NRF Services

The LMF implements the NF service consumer for the `Nnrf_NFManagement` and `Nnrf_NFDiscovery` services per *TS 29.510*.

Table 3 SBIs for NRF Services

Service Name	Service Operations	Operation Semantics
Nnrf_NFManagement	NFRegister	Request/Response

Service Name	Service Operations	Operation Semantics
	NFUpdate	
	NFDeregister	
	NFStatusSubscribe	Subscribe/Notify
	NFStatusNotify	
	NFStatusUnsubscribe	
Nnrf_NFDiscovery	Request	Request/Response
Nnrf_AccessToken	AccessTokenRequest	Request/Response

The LMF supports TLS encryption for security protection at the transport layer of SBIs. The LMF supports both server-side and client-side certificates for TLS authentication. TLS 1.2 and TLS 1.3 are supported.

3 Operation and Maintenance

3.1 Alarms

The NLS supports the following alarms for the LMF:

— Internal Communication for LMF

— Network Connection to 5G Core

— License for LMF Value Package

— Other alarms related to positioning methods

For more details about the alarms, see the Related Information.

—— RELATED INFORMATION ——

Alarm Descriptions, NLS

3.2 Statistics

The NLS generates statistics counters for LMF positioning. The counters record the number of location requests from the AMF for different client types and positioning methods, the average response time, and so on. In the NLS, many statistics counters can be split on system level, cell level, and client type level.

For more details, see the Related Information.

—— RELATED INFORMATION ——

Statistics Description, NLS

3.3 Positioning Records

For the LMF, the NLS supports XML-based positioning records for the location requests, including additional elements for the NR network.

Note: The data types of the location data in the NR network are according to *3GPP 29.572*. These data types are different from other network access types.

For more details, see the Related Information.

—— RELATED INFORMATION ——

Positioning Record File Description, NLS

3.4 Configuration Management

The configuration of the LMF feature includes the following:

- Component parameters on NLS components
- Positioning control data configuration, for the positioning method selection and fallback

For more details, see the Related Information.

—— RELATED INFORMATION ——

Parameter List, NLS

Positioning Control Data File and CLI Description

3.5 Cell Data Management

To use LMF positioning, the NLS must be aware of the geographic location of the NR cells. The NR cell data can be imported from an XML file, provisioned through CLI or the Administration Portal.

For more details, see the Related Information.

—— RELATED INFORMATION ——

NR Cell Data File and CLI Description

3.6 License Handling

The following licenses are needed for the LMF in the NLSs:

- To support 5G LMF commercial positioning, the Value Package (VP) licenses for LMF must be installed. To use A-GNSS positioning method, the VP licenses for AGNSS must be installed.
- To support 5G LMF emergency positioning, the regulatory VP licenses for LMF and NLS Mobile Emergency must be installed in the NLS.

Note: The maximum number of NR cells must not exceed the cell quantity of the VP licenses for LMF.

For more details on licenses, see the Related Information.

—— RELATED INFORMATION ——

Technical Product Description, NLS 2.4

3.7 Administration Portal

The Administration Portal provides a web-based interface to manage the LMF value package. It includes the following functions:

- Positioning method control for the LMF
- Data management for:
  - Cell data
  - Cells report
  - Fixed Ghost data for the NR network
- Software management for the LMF-related components
- System record
  - Statistics
  - Positioning record
- System monitor
  - GhostAlarm log
  - Active Alarm list

For details, see the Related Information.

—— RELATED INFORMATION ——

Administration Portal User Guide for ENL Cloud Native (NLS)

3.8 High Availability

The NLS supports dual-server cluster configuration for LMF high availability. The static application data, the NR cell data, and the positioning control data, for example, are stored in a database with redundancy.

The LMF services are active on both NLS servers. The LMF service on each NLS server registers an NF instance in the NRF with a different NF instance ID. The NRF servers are redundant. The LMF instance sends `PING` messages to the NRF servers, and handles failovers when any of the NRF servers fail. During a failover, the LMF instance is registered again to an available NRF server. A missing or faulty heartbeat response also results in failover.

During service discovery, the AMF gets multiple LMF NF instances from the NRF. The AMF communicates with the selected LMF instances directly and handles failovers when any of the LMF servers fail.

3.9 5G LMF Proxy Service

The 5G LMF Proxy can be used as a load balancer service if it is deployed and configured as the proxy service. In this case, the following TLS settings depend on the configuration of the proxy:

- HTTP/HTTPS used in the service address that is included in the NF profile
- Client connections towards the AMF

If the 5G LMF Proxy service is enabled, multiple LMF Pod instances are supported.

Deploying the proxy is not mandatory if there is only one LMF TS Pod, or if the only used positioning method is CID. Otherwise, the proxy service must be configured as the service address of the LMF.

To enable the 5G LMF Proxy service:

Steps

1. Set the value of `PROXY_SERVICE_NAME` to **`enl-5g-proxy`**.
2. Enable TLS by setting the value of `useTLS` to **`true`** under `services` in the `values.yaml` file during the installation.

Note:

— If the `enl-5g-proxy` Pod is used, TLS can only be configured during the deployment by using the above method.

— If the 5G LMF Proxy service is not used, and the specified proxy service is `nls-ts-lmf`, TLS can also be turned on and off runtime by enabling the `LMF_Server_TLS_Switch` server level configuration parameter in **Software Management** > **NLS Components** > **Configuration** > **PPLMF** in the Administration Portal.

For more information on setting the value of `PROXY_SERVICE_NAME`, see the section mentioned in the Related Information.

— If the deployment is instantiated with the Evolved Virtual Network Function Manager (EVNFM), then, instead of changing the Helm chart, the attributes must be set on the EVNFM GUI or in the input `values.yaml` file. For more information, see the *Instantiate a VNF* section in *EVNFM User Guide* in the Ericsson Orchestrator EVNFM CPI library, version 22.2.3.

3. Set the address of the load balancer service; as described in Load Balancing with 5G LMF Proxy Service on page 49.

—— RELATED INFORMATION ——

NLS Configuration, Installation Instruction

3.9.1 Load Balancing with 5G LMF Proxy Service

If the 5G LMF Proxy service is active, then the service acts as a load balancer. Sessions are identified by the LCS correlation ID.

The IP address and the port of the proxy service are included in the NF profile as the service address. The address can be configured by one of the following methods:

— To configure the address during the installation, set the value of `externalAddress` to the address of the load balancer in the `values.yaml` file.

— To configure the address runtime, set the value of the `Load_Balancer_URL` PPLMF cluster level configuration parameter to the URL of the load balancer in **Software Management** > **NLS Components** > **Configuration** > **PPLMF** in the Administration Portal.

When providing the address, the following formats can be used:

— `https://IP:port`

- `http://IP:port`

Note:

- When setting the address runtime, the PPLMF component must be restarted so that it can register to the NRF with the new service profile and the new address.
- If the deployment is instantiated with the EVNFM, then the `externalAddress` to the load balancer must be set during the instantiation process of the EVNFM either on the GUI or in the input `values.yaml` file. For more information, see the *Instantiate a VNF* section in *EVNFM User Guide* in the Ericsson Orchestrator EVNFM CPI library, version 22.2.3.

4 Engineering Guidelines

In connection with the 5G LMF feature, the following engineering guidelines must be considered:

— The 5G software design is based on the assumption that the product will be integrated into the network using the mTLS as mandated by the 3GPP standard. This ensures mutual authentication between the network elements, provides integrity protection for messages and encrypts the data stream, which may contain sensitive data. In this regard, there is no other control built into the ENL in case of disabled mTLS.

  Disabling mTLS is only recommended after a comprehensive risk analysis, taking into account the possible technical and business impacts.

— For TLS, Ericsson strongly recommends installing certificates issued by a trusted certificate authority (CA) instead of using self-signed certificates. It is assumed that the certificates of all AMF instances can be verified using the same root certificate. For information on how to install customized certificates, see *Create and Setup a CA-Signed Certificate* section mentioned in the Related Information.

— The LMF follows the same design principles as the E-SMLC in the NLS whenever possible. However, there are the following differences and limitations in the current LMF release:

  - The positioning method control configuration is supported only on global level.
  - Dual technology and hybrid methods are not supported.

—— RELATED INFORMATION ——

Create and Setup a CA-Signed Certificate